1. Write a program in Python to accept an integer value in variable N from the user and display its content in the following patterns.

Sample output:

```
Enter Integer: 5
1
12
123
1234
12345
```

```
54321
4321
321
21
1
```

```
    1
   121
  12321
 1234321
123454321
```

2. Write a program in Python to accept an integer value in variable N from the user and display its content in the following patterns.

Sample output:

```
Enter Integer: 5
1
22
333
4444
55555
```

```
    1
   22
  333
 4444
55555
```

```
    1
   222
  33333
 4444444
555555555
```

3. Write a program in Python to accept a string from a user and display its content in the following patterns.

Sample output:

```
Enter String: GOGREEN
G
GO
GOG
GOGR
GOGRE
GOGREE
GOGREEN
```

```
GOGREEN
GOGREE
GOGRE
GOGR
GOG
GO
G
```

```
      G
     GO
    GOG
   GOGR
  GOGRE
 GOGREE
GOGREEN
```

4. Write a program in Python to accept a string from a user and display its content in the following patterns.

Sample output:

```
Enter String: WATER
W
 A
  T
   E
    R
```

```
    R
   E
  T
 A
W
```

```
W       W
 A     A
  T   T
   E E
    R
```

5. Write a program in Python to accept a string from the user and show only non vowels (display underscore _ symbol in place of vowel).

Sample output:

```
Enter String: WATER
W_T_R
```

```
Enter String: PEACE
P_ _C_
```

6. Write a program in Python to accept a word WRD and a sentence STC as strings from the user. Find the presence or absence of the word WRD in the sentence STC (Without using **`if WRD in STC`**).

Sample output:

```
Sentence: A small box in the big box
Word: box
Status: Found
```

7. Write a program in Python to accept a sentence as a string from the user. Find the total number of words present in the string (Without using **`in`** operator with **`if`**)

Sample output:

```
Sentence: A small box in the big box
Total Number of Words: 7
```

8. Write a program in Python to accept a word WRD and a sentence STC as strings from the user. Find the positions of each occurrence of word WRD in the sentence STC. (Can use split() function)

Sample output:

```
Sentence: A small box in the big box
Word: box
Positions: 8, 23,
```

9. Write a program in Python to accept a string S from a user and display an encoded content of S containing the reverse of the first 3 characters and last 3 characters.

Sample output:

```
Enter S: HALF OF THE TEAM
Coded S: LAHEAM
```

```
Enter S: MOHAN'S TEAM SCORED 345
Coded S: HOM345
```

10. Write a program in Python to accept a string S from a user and display an encoded content of S containing the reverse of characters of first half of S and original order of characters of second half of the S.

Sample output:

```
Enter S: ABCDEFGHI
Coded S: EFGHIDCBA
```

```
Enter S: ABCDEFGH
Coded S: EFGHDCBA
```

11. Write a program in Python to accept a string from a user and display the count of occurrence of vowels, digits and consonants in it.

Sample output:

```
Enter Message: SAVE WATER
VOWELS: 4
CONSONANTS: 5
DIGITS: 0
```

```
Enter Message: 10 HAFTE 10 BAJE
VOWELS: 4
CONSONANTS: 5
DIGITS: 4
```

12. Write a program in Python to accept a string from a user and check if it is Palindrome or not.

Sample output:

```
Enter String: NITIN
It is a PALINDROME
```

```
Enter String: DELHI
It is not a PALINDROME
```

13. Write a Python code to create a two player word guessing game with the help of following steps.
    1. Accept a string **Str** from User1 [accept only those strings, which are less than 10 alphabets]
    2. Scroll the screen with the help of print statements to hide the User1's string.

3. Display the content of string by showing consonants' position as underscore _ and vowels as it is. Name it as **GuessedStr**
4. Accept a single alphabet from User2
5. If alphabet entered by the User2 matches with one of the alphabets of User1's string, re-display the GuessStr with the alphabet at appropriate place
6. Repeat steps 4 and 5 till the user consumes his 10 chances of guessing all the correct letters of **Str**.
7. At the end display score as 20 - No. of attempts by User2. Deduct 2 points for every re-input of the same alphabet again.

Sample output:

```
User1 String: WATER CONSERVATION
Re-Enter [Not more than 10 letters]
User1 String: WATER
:
:
Hint: _A_E_
User2 Guess [1]: U
** U not matching with any
User2 Guess [2]: W
Hint: WA_E_
User2 Guess [3]: K
** K not matching with any
User2 Guess [4]: W
** You lose 2 points for re-entering the same letter
User2 Guess [5]: R
Hint: WA_ER
User2 Guess [6]: T
Great - Well done, you guessed right, it is WATER
Your Score: 12
[i.e. Full score 20 - Penalty 2 - Number of Times 6]
```

IMPORTANT: REFER TO MORE QUESTIONS GIVEN IN ASSIGNMENT BOOKLET

General Instructions:

i. Type the solutions of all the problems using **Python** Language on **Python IDLE** or **colab.research.google.com**

ii. Type in the following on top of each of your program as comment lines

```
# ------------------------------------------------------------#
# List-Program No     : L3-P1
# Developed By        : Sukrit Joy
# Date                : 17-Aug-2023
# ------------------------------------------------------------#
```

iii. On successful execution with meaningful data, **copy** and **paste** the sample output at the bottom of the program as comment lines.

iv. "**Turn in**" the softcopy of the programs as a single document in response to this assignment submission with each program on separate pages. Once verified by the teacher, take out a hardcopy of **each program on separate pages** from the printer and get them signed by the respective computer teacher. Add all the printouts in the practical file and get the Index entry also signed.

v. Recommended Font, "**Courier New**", Style "**Bold**" Size "**10**" for all programs with single line spacing. Default indentation **2 Character** only.

```
# ------------------------------------------------------------#
# List-Program No    : L3-P1
# Developed By       : PUJA DEY
# Date               : 16-August-2024
# ------------------------------------------------------------#

N=int(input("Enter a no."))
for C in range(1,N+1):
  for R in range(1,C+1):
    print(R,end="")
  print()
```

```
Enter a no.5
1
12
123
1234
12345
```

```
N=int(input("Enter a no."))
for C in range(N,0,-1):
  for R in range(C,0,-1):
    print(R,end="")
  print()
```

```
Enter a no.5
54321
4321
321
21
1
```

```
N=int(input("Enter a no."))
for I in range(1,N+1):
  print(end=(5-I)*" ")
  for J in range(1,I+1):
    print(J,end="")
  for J in range(I-1,0,-1):
    print(J,end="")
  print()
```

```
Enter a no.5
    1
   121
  12321
 1234321
123454321
```

```
# ------------------------------------------------------------#
# List-Program No    : L3-P2
# Developed By       : PUJA DEY
# Date               : 16-August-2024
# ------------------------------------------------------------#
N=int(input("Enter a no."))
for C in range(1,N+1):
  for R in range(1,C+1):
    print(C,end="")
  print()
```

```
Enter a no.5
1
22
333
4444
55555
```

```
N=int(input("Enter a no."))
for I in range(1,6):
  for J in range(1,6-I):
    print(end=" ")
  for J in range(1,I+1):
    print(I, end="")
  print()
```

```
Enter a no.5
    1
   22
  333
 4444
55555
```

```
N=int(input("Enter a no."))
for I in range(1,N+1):
  for J in range(1,6-I):
    print(end=" ")
  for J in range(1,2*I):
    print(I,end="")
  for J in range(1,6-I):
    print(end=" ")
  print()
```

```
Enter a no.5
    1
   222
  33333
 4444444
555555555
```

```
# ------------------------------------------------------------#
# List-Program No    : L3-P3
# Developed By       : PUJA DEY
# Date               : 16-August-2024
# ------------------------------------------------------------#
```

```
S=str(input("Enter a string "))
L=len(S)
for C in range(L):
  for R in range(0,C+1):
    print(S[R],end="")
  print()
```

```
Enter a string GOGREEN
G
GO
GOG
GOGR
GOGRE
GOGREE
GOGREEN
```

```
S=str(input("Enter a string "))
L=len(S)
for C in range(L,0,-1):
  for J in range(C):
    print(S[J], end="")
  print()
```

```
Enter a string GOGREEN
GOGREEN
GOGREE
GOGRE
GOGR
GOG
GO
G
```

```
S=str(input("Enter a string "))
L=len(S)
for C in range(L):
  for J in range(0,L-C):
    print(end=" ")
  for J in range(0,C+1):
    print(S[J], end="")
  print()
```

```
Enter a string GOGREEN
       G
      GO
     GOG
    GOGR
   GOGRE
  GOGREE
 GOGREEN
```

```
# ------------------------------------------------------------#
# List-Program No    : L3-P4
# Developed By       : PUJA DEY
# Date               : 16-August-2024
# ------------------------------------------------------------#

S=str(input("Enter a string: "))
L=len(S)
for C in range(L):
  print(" "*C, S[C] )
```

```
Enter a string: WATER
 W
  A
   T
    E
     R
```

```
S=str(input("Enter a string: "))
for i in range(len(S)-1,-1,-1):
    print(' ' * (i) + S[i])
```

```
Enter a string: WATER
    R
   E
  T
 A
W
```

```
S=str(input("Enter a string: "))
L=len(S)
for i in range(L):
   print(' ' * i + S[i] + ' ' *(2*(L - i - 1)-1) + (S[i] if i != L - 1 else ''))
```

```
Enter a string: WATER
W       W
 A     A
  T   T
   E E
    R
```

```
# ------------------------------------------------------------#
# List-Program No    : L3-P5
# Developed By       : PUJA DEY
# Date               : 16-August-2024
# ------------------------------------------------------------#

while True:
  S=str(input("Enter a string "))
  S1=S.replace("A", "_")
  S2=S1.replace("E", "_")
  S3=S2.replace("I", "_")
  S4=S3.replace("O", "_")
  S5=S4.replace("U", "_")
  print(S5)
  ch=input("Want for more strings? (Y/N): ")
  if ch=="N":
   break
```

```
Enter a string WATER
W_T_R
Want for more strings? (Y/N): Y
Enter a string PEACE
P__C_
Want for more strings? (Y/N): N
```

```
# ------------------------------------------------------------#
# List-Program No    : L3-P6
# Developed By       : PUJA DEY
# Date               : 16-August-2024
# ------------------------------------------------------------#

STC = input("Enter a Sentence: ")
WRD = input("Enter a word: ")
A = STC.split()
for i in A:
  if i == WRD:
    print("Status: Found")
    break
else:
  print("Status: Not Found")
```

```
Enter a Sentence: A big box in a small box
Enter a word: box
Status: Found
```

```
# ------------------------------------------------------------#
# List-Program No    : L3-P7
# Developed By       : PUJA DEY
# Date               : 16-August-2024
# ------------------------------------------------------------#

def count_words(sentence):
  words=sentence.split()
  return len(words)

STC=str(input("Enter the Sentence "))
total_words=count_words(STC)
print("No of words are:",total_words)
```

```
Enter the Sentence A small box in the big box
No of words are: 7
```

```
# ------------------------------------------------------------#
# List-Program No    : L3-P8
# Developed By       : PUJA DEY
# Date               : 16-August-2024
# ------------------------------------------------------------#

def find_word_positions(sentence, word):
    words = sentence.split()
    positions = []
    index = 0
    for w in words:
        if w == word:
            positions.append(index)
        index += len(w) + 1
    return positions

STC = input("Enter a sentence: ")
WRD = input("Enter the word to find: ")
positions = find_word_positions(STC, WRD)

if positions:
    position_str = ""
    for i in range(len(positions)):
        position_str += str(positions[i])
        if i < len(positions) - 1:
            position_str += ", "
    print("Positions: " + position_str + ",")
else:
    print("Word not found.")
```

```
Enter a sentence: A small box in the big box
Enter the word to find: box
Positions: 8, 23,
```

```
# ------------------------------------------------------------#
# List-Program No    : L3-P9
# Developed By       : PUJA DEY
# Date               : 16-August-2024
# ------------------------------------------------------------#

Str=str(input("Enter the Sentence "))
print(Str[2::-1], end="")
Str1=Str[-1:-4:-1]
print(Str1[::-1])
```

```
Enter the Sentence MOHAN'S TEAM SCORED 345
HOM345
```

```
# -------------------------------------------------------------#
# List-Program No    : L3-P10
# Developed By       : PUJA DEY
# Date               : 16-August-2024
# -------------------------------------------------------------#

Str=str(input("Enter the Sentence "))
L=len(Str)
Str1=Str[L//2::1]
Str2=Str[0:L//2]
Str3=Str2[::-1]
print(Str1+Str3)
```

```
Enter the Sentence ABCDEFGH
EFGHDCBA
```

```
# ------------------------------------------------------------#
# List-Program No    : L3-P11
# Developed By       : PUJA DEY
# Date               : 16-August-2024
# ------------------------------------------------------------#

Str=str(input("Enter the Sentence "))
Vowels="AEIOUaeiou"
Vowels_Count=0
Digits_Count=0
Consonants_Count=0

Str1=Str.replace(" ","")
for char in Str1:
  if char in Vowels:
    Vowels_Count=Vowels_Count+1
  elif char.isnumeric():
    Digits_Count=Digits_Count+1
  elif char.isalpha():
    Consonants_Count+=1

print("No of vowels", Vowels_Count)
print("No of Digits", Digits_Count)
print("No of Consonants", Consonants_Count)
```

```
Enter the Sentence 10 HAFTE 10 BAJE
No of vowels 4
No of Digits 4
No of Consonants 5
```

```
# -----------------------------------------------------------------#
# List-Program No    : L3-P12
# Developed By       : PUJA DEY
# Date               : 16-August-2024
# -----------------------------------------------------------------#

st = input("Enter a String: ")
if st==st[::-1]:
  print("The given string",st,"is a palindrome")
else:
  print("The given string",st,"is not a palindrome")
```

```
Enter a String: NITIN
The given string NITIN is a palindrome
```

```
# ------------------------------------------------------------#
# List-Program No    : L3-P13
# Developed By       : PUJA DEY
# Date               : 16-August-2024
# ------------------------------------------------------------#

user1_string = input("User1 String: ")

while len(user1_string) > 10:
        print("Please enter a string with no more than 10 letters.")
        user1_string = input("Re-Enter [Not more than 10 letters]: ")

print("\n" * 50)

guessed_str = ''
for ch in user1_string:
        if ch in 'AEIOU':
            guessed_str += ch
        else:
            guessed_str += '_'
print("Hint:", guessed_str)

attempts_left = 10
guessed_letters = []
score = 20
count=0

for attempt in range(1, attempts_left + 1):
        guess = input("User2 Guess [" + str(attempt) + "]: ")

        if guess in guessed_letters:
            print("** You lose 2 points for re-entering the same letter")
            score -= 4
            count=count+1
        else:
            guessed_letters.append(guess)

            if guess in user1_string:
                new_guessed_str = ''
                for i in range(len(user1_string)):
```

```
                    if user1_string[i] == guess:
                        new_guessed_str += guess
                    else:
                        new_guessed_str += guessed_str[i]
                guessed_str = new_guessed_str

                print("Hint:", guessed_str)

                if guessed_str == user1_string:
                    print("Great - Well done, you guessed right, it is " +
user1_string)
                    break
            else:
                print("** " + guess + " not matching with any")

            if guessed_str != user1_string:
                score -= 1

print("Your Score:", score)
print("[i.e. Full score 20 - Penalty 2*" , count, "- Number of Times", attempt ,
"]")
```

```
User1 String: WATER CONSERVATION
Please enter a string with no more than 10 letters.
Re-Enter [Not more than 10 letters]: WATER
```

```
Hint: _A_E_
User2 Guess [1]: U
** U not matching with any
User2 Guess [2]: W
Hint: WA_E_
User2 Guess [3]: K
** K not matching with any
User2 Guess [4]: W
** You lose 2 points for re-entering the same letter
User2 Guess [5]: T
Hint: WATE_
User2 Guess [6]: R
Hint: WATER
Great - Well done, you guessed right, it is WATER
Your Score: 12
[i.e. Full score 20 - Penalty 2* 1 - Number of Times 6 ]
```